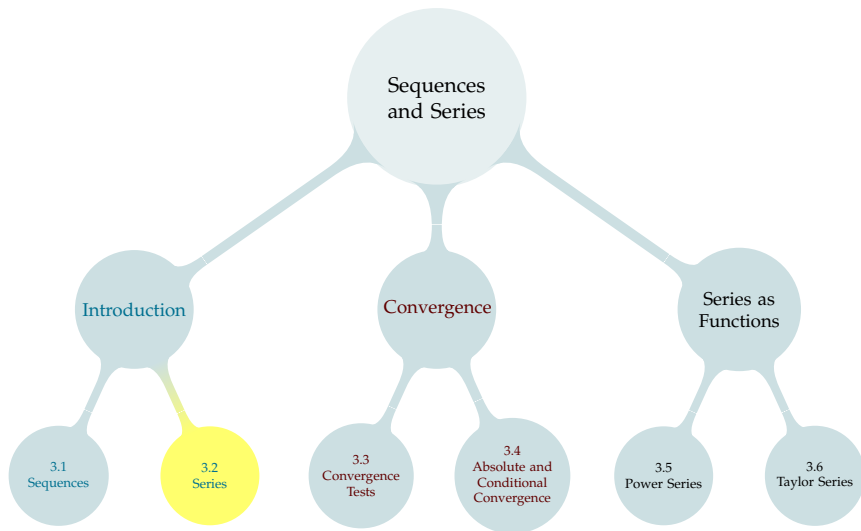


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SEQUENCES AND SERIES

A **sequence** is a list of numbers

A **series** is the sum of such a list.

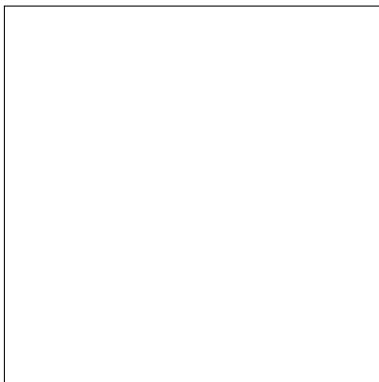
SEQUENCES AND SERIES

Sequence

List of numbers,
approaching

Series

Sum of numbers,
approaching



Square of Area 1

QUICK REVIEW: SIGMA NOTATION

Recall:

$$\sum_{n=1}^5 \frac{1}{n^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2}$$

We informally interpret:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2} + \frac{1}{8^2} + \frac{1}{9^2} + \frac{1}{10^2} + \cdots$$

(a more rigorous definition will be discussed soon)

Let a_n and b_n be sequences, and let C be a constant.

$$\sum_{n=1}^{\infty} (C \cdot a_n) =$$

A. $\sum_{n=1}^{\infty} C \cdot \sum_{n=1}^{\infty} a_n$

B. $\sum_{n=1}^{\infty} C + \sum_{n=1}^{\infty} a_n$

C. $C \sum_{n=1}^{\infty} a_n$

D. $a_n \sum_{n=1}^{\infty} C$

E. none of the above

Let a_n and b_n be sequences, and let C be a constant.

$$\sum_{n=1}^{\infty} (a_n + b_n) =$$

A. $\sum_{n=1}^{\infty} a_n \cdot \sum_{n=1}^{\infty} b_n$

B. $\sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$

C. $a_n + \sum_{n=1}^{\infty} b_n$

D. $a_n \sum_{n=1}^{\infty} b_n$

E. none of the above

Let a_n and b_n be sequences, and let C be a constant.

$$\sum_{n=1}^{\infty} (a_n)^C =$$

A. $\sum_{n=1}^{\infty} C \cdot \sum_{n=1}^{\infty} a_n$

B. $\left(\sum_{n=1}^{\infty} a_n \right)^C$

C. $C^n \sum_{n=1}^{\infty} a_n$

D. $\sum_{n=1}^{\infty} C(a_n)^{C-1}$

E. none of the above



SERIES PHILOSOPHY

What does it really mean to add up infinitely many things?

$$1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + \cdots$$

We need an unambiguous definition.

HOW CAN WE ADD UP INFINITELY MANY THINGS?

SEQUENCE OF PARTIAL SUMS



$$\frac{1}{5^1}$$



$$\frac{1}{5^2}$$



$$\frac{1}{5^3}$$



$$\frac{1}{5^4}$$



$$\frac{1}{5^5}$$

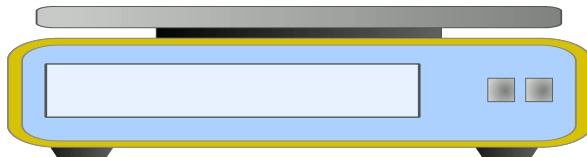
$$S_1 = 0.2000$$

$$S_2 = 0.2400$$

$$S_3 = 0.2480$$

$$S_4 = 0.2496$$

$$S_5 = 0.2499$$



Partial sums let us think about series (sums) using the tools we've developed for sequences (lists).

$$a_1 = \frac{1}{5} = 0.2$$

$$S_1 = 0.2$$

$$a_2 = \frac{1}{5^2} = 0.04$$

$$S_2 = 0.24$$

$$a_3 = \frac{1}{5^3} = 0.008$$

$$S_3 = 0.248$$

$$a_4 = \frac{1}{5^4} = 0.0016$$

$$S_4 = 0.2496$$

$$a_5 = \frac{1}{5^5} = 0.00032$$

$$S_5 = 0.24992$$

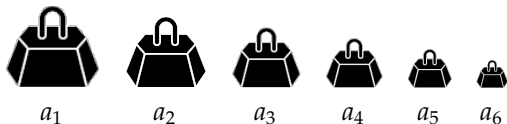
We define $\sum_{n=1}^{\infty} a_n = \lim_{N \rightarrow \infty} \sum_{n=1}^N a_n = \lim_{N \rightarrow \infty} S_N$.

$a_1 = \frac{1}{5} = 0.2$	$S_1 = 0.2$	$a_5 = \frac{1}{5^5} = 0.00032$	$S_5 = 0.24992$
$a_2 = \frac{1}{5^2} = 0.04$	$S_2 = 0.24$	$a_6 = \frac{1}{5^6} = 0.000064$	$S_6 = 0.249984$
$a_3 = \frac{1}{5^3} = 0.008$	$S_3 = 0.248$	$a_7 = \frac{1}{5^7} = 0.0000128$	$S_7 = 0.2499968$
$a_4 = \frac{1}{5^4} = 0.0016$	$S_4 = 0.2496$	$a_8 = \frac{1}{5^8} = 0.00000256$	$S_8 = 0.24999936$

From the sequence of partial sums, we guess

$$\sum_{n=1}^{\infty} = \lim_{N \rightarrow \infty} S_N =$$

NOTATION: $S_N = \sum_{n=1}^N a_n$



$$S_1 = a_1$$

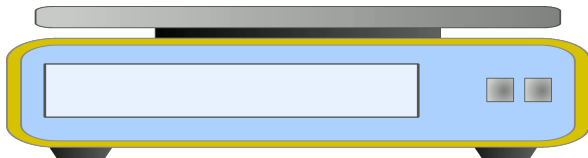
$$S_2 = a_1 + a_2$$

$$S_3 = a_1 + a_2 + a_3$$

$$S_4 = a_1 + \cdots + a_4$$

$$S_5 = a_1 + \cdots + a_5$$

$$S_6 = a_1 + \cdots + a_6$$



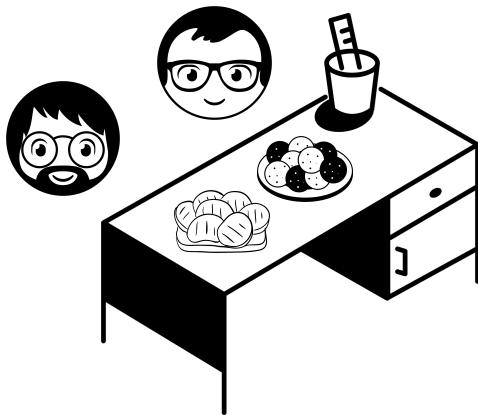
NOTATION PRACTICE

Suppose $\sum_{n=1}^{\infty} a_n$ has partial sums $S_N = \sum_{n=1}^N a_n = \frac{N}{N+1}$.

► Evaluate $\sum_{n=1}^{100} a_n$.

► Evaluate $\sum_{n=1}^{\infty} a_n$.

NOTATION PRACTICE

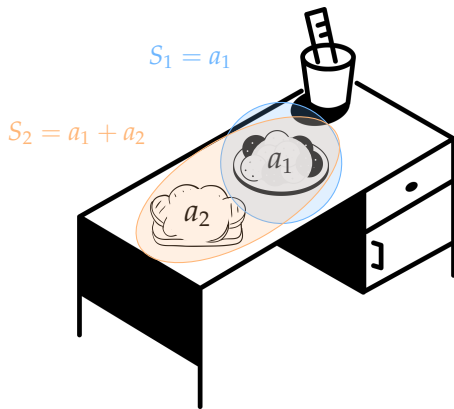


Andrew brings a plate of cookies to the professor's desk. When he puts them down, there are 10 cookies on the desk.

Then, Joel brings a plate of cookies. When he puts them down, there are 19 cookies on the desk.

How many cookies did each person bring?

NOTATION PRACTICE



Andrew brings a plate of cookies to the professor's desk. When he puts them down, there are 10 cookies on the desk.

Then, Joel brings a plate of cookies. When he puts them down, there are 19 cookies on the desk.

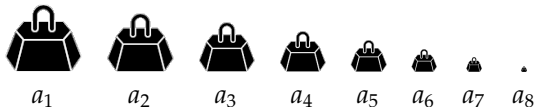
How many cookies did each person bring?

NOTATION PRACTICE

Suppose $\sum_{n=1}^{\infty} a_n$ has partial sums $S_N = \sum_{n=1}^N a_n = \frac{N}{N+1}$.

- Find a_1 .
- Give an explicit expression for a_n , when $n > 1$.

$$S_N = \sum_{n=1}^N a_n = \frac{N}{N+1}$$



$$S_1 = 1/(1+1)$$

$$S_2 = 2/(2+1)$$

$$S_3 = 3/(3+1)$$

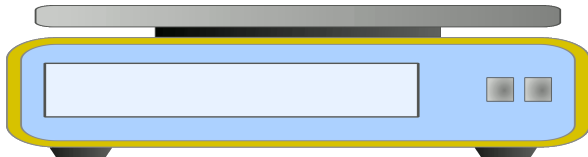
$$S_4 = 4/(4+1)$$

$$S_5 = 5/(5+1)$$

$$S_6 = 6/(6+1)$$

$$S_7 = 7/(7+1)$$

$$S_8 = 8/(8+1)$$



Definition

The N^{th} partial sum of the series $\sum_{n=1}^{\infty} a_n$ is the sum of its first N terms

$$S_N = \sum_{n=1}^N a_n.$$

The partial sums form a sequence $\{S_N\}_{N=1}^{\infty}$. If this sequence of partial sums converges $S_N \rightarrow S$ as $N \rightarrow \infty$ then we say that the series $\sum_{n=1}^{\infty} a_n$ converges to S and we write

$$\sum_{n=1}^{\infty} a_n = S$$

If the sequence of partial sums diverges, we say that the series diverges.

Geometric Series

Let a and r be two fixed real numbers with $a \neq 0$. The series

$$a + ar + ar^2 + ar^3 + \dots$$

is called the **geometric series** with first term a and ratio r .

We call r the *ratio* because it is the quotient of consecutive terms:

$$\frac{ar^{n+1}}{ar^n} = r$$

Another useful way of identifying geometric series is to determine whether all pairs of consecutive terms have the same ratio.

- ▶ Geometric: $1 + \frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \frac{1}{5^4} + \dots$
- ▶ Geometric: $\sum_{n=0}^{\infty} \frac{1}{2^n}$
- ▶ Not geometric: $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \dots$

Consider the partial sum S_N of a geometric series:

$$S_N = a + ar + ar^2 + ar^3 + \cdots + ar^N$$

$$rS_N =$$

$$rS_N - S_N =$$

$$S_N(r - 1) = ar^{N+1} - a$$

If $r \neq 1$, then

$$S_N = \frac{ar^{N+1} - a}{r - 1} = a \frac{r^{N+1} - 1}{r - 1}$$

Geometric Series and Partial Sums

Let a and r be constants with $a \neq 0$, and let N be a natural number.

► If $r \neq 1$, then $a + ar + ar^2 + ar^3 + \cdots + ar^N = a \frac{r^{N+1} - 1}{r - 1}$.

► If $r = 1$, then $a + ar + ar^2 + ar^3 + \cdots + ar^N =$

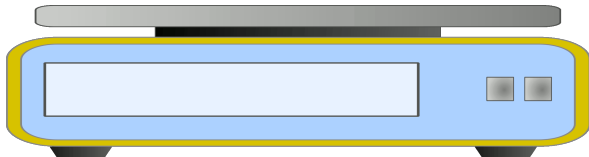
► If $|r| < 1$, then $\sum_{n=0}^{\infty} ar^n =$

► If $r = 1$, then $\sum_{n=0}^{\infty} ar^n$

► If $r = -1$, then $\sum_{n=0}^{\infty} ar^n$

► If $|r| > 1$, then $\sum_{n=0}^{\infty} ar^n$

$$\sum_{n=0}^{\infty} ar^n, r = 1, a \neq 0$$



$$S_0 = a$$

$$S_1 = 2a$$

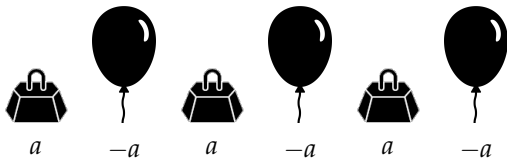
$$S_2 = 3a$$

$$S_3 = 4a$$

$$S_4 = 5a$$

$$S_5 = 6a$$

$$\sum_{n=0}^{\infty} ar^n, r = -1, a \neq 0$$



$$S_0 = a$$

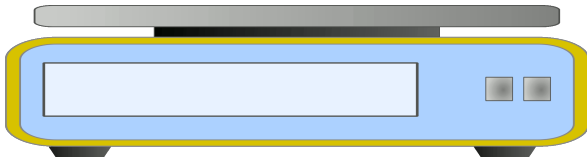
$$S_1 = 0$$

$$S_2 = a$$

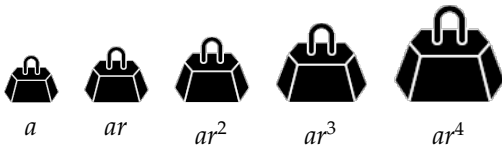
$$S_3 = 0$$

$$S_4 = a$$

$$S_5 = 0$$



$$\sum_{n=0}^{\infty} ar^n, r > 1, a \neq 0$$



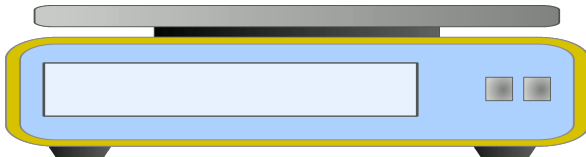
$$S_0 = a$$

$$S_1 = a \frac{r^2 - 1}{r - 1}$$

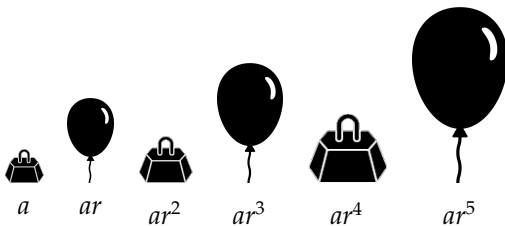
$$S_2 = a \frac{r^3 - 1}{r - 1}$$

$$S_3 = a \frac{r^4 - 1}{r - 1}$$

$$S_4 = a \frac{r^5 - 1}{r - 1}$$



$$\sum_{n=0}^{\infty} ar^n, r < -1, a \neq 0$$



$$S_0 = a$$

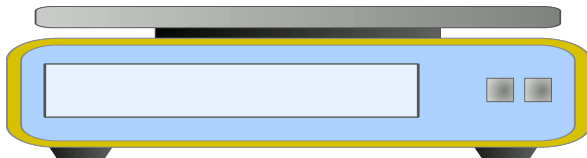
$$S_1 = a \frac{r^2 - 1}{r - 1}$$

$$S_2 = a \frac{r^3 - 1}{r - 1}$$

$$S_3 = a \frac{r^4 - 1}{r - 1}$$

$$S_4 = a \frac{r^5 - 1}{r - 1}$$

$$S_5 = a \frac{r^6 - 1}{r - 1}$$



GEOMETRIC SERIES

New bitcoins are produced when a particular type of computational problem is solved. Every time 210,000 solutions are found, the number of bitcoins each solution can produce is cut in half.

- ▶ Each of the first 210,000 solutions can produce 50 bitcoins.
- ▶ Each of the next 210,000 solutions can produce $\frac{50}{2}$ bitcoins.
- ▶ Each of the next 210,000 solutions can produce $\frac{50}{2^2}$ bitcoins.
- ▶ Each of the next 210,000 solutions can produce $\frac{50}{2^3}$ bitcoins.

Assume that this continues forever, and that bitcoins are infinitely divisible.¹ How many bitcoins can possibly be produced?

¹Actually the smallest allowed division of a bitcoin is 10^{-8} .

$$\sum_{n=0}^{\infty} 210\,000 \left(\frac{50}{2^n} \right) = 21\,000\,000$$



10 500 000



5 250 000



2 625 000



1 312 500



656 250

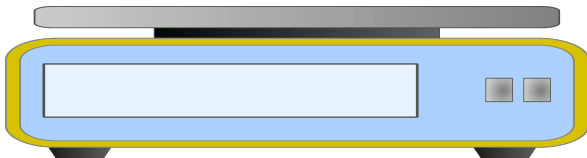
$$S_0 = 10\,500\,000$$

$$S_1 = 15\,750\,000$$

$$S_2 = 18\,375\,000$$

$$S_3 = 19\,687\,500$$

$$S_4 = 20\,343\,750$$



Arithmetic of Series

Let S , T , and C be real numbers. Let the two series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge to S and T respectively. Then

$$\sum_{n=1}^{\infty} [a_n + b_n] = S + T$$

$$\sum_{n=1}^{\infty} [a_n - b_n] = S - T$$

$$\sum_{n=1}^{\infty} [Ca_n] = CS$$

Geometric Series and Partial Sums

Let a and r be fixed numbers, and let N be a positive integer. Then

$$\sum_{n=0}^N ar^n = \begin{cases} a \cdot \frac{1-r^{N+1}}{1-r} & \text{if } r \neq 1 \\ a(N+1) & \text{if } r = 1 \end{cases}$$

so

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \text{ provided } |r| < 1$$

Evaluate $\sum_{n=0}^{\infty} \left(\frac{2}{3^n} + \frac{4}{5^n} \right)$

Geometric Series and Partial Sums

Let a and r be fixed numbers, and let N be a positive integer. Then

$$\sum_{n=0}^N ar^n = \begin{cases} a \cdot \frac{1-r^{N+1}}{1-r} & \text{if } r \neq 1 \\ a(N+1) & \text{if } r = 1 \end{cases}$$

so

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \text{ provided } |r| < 1$$

Evaluate $\sum_{n=6}^{\infty} \left(\frac{3^{n-1}}{5^{2n}} \right)$

Geometric Series and Partial Sums

Let a and r be fixed numbers, and let N be a positive integer. Then

$$\sum_{n=0}^N ar^n = \begin{cases} a \cdot \frac{1-r^{N+1}}{1-r} & \text{if } r \neq 1 \\ a(N+1) & \text{if } r = 1 \end{cases}$$

so

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \text{ provided } |r| < 1$$

Evaluate $\sum_{n=0}^{\infty} \left(\frac{2^{2n}}{3^n} \right)$



TELESCOPING SUMS

Evaluate $\sum_{n=1}^{800} \left(\frac{1}{n} - \frac{1}{n+1} \right)$.

Evaluate $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$.

Evaluate $\sum_{n=1}^{1000} \log \left(\frac{n+1}{n} \right) .$

Evaluate $\sum_{n=1}^{\infty} \log \left(\frac{n+1}{n} \right) .$

Included Work

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